ASSET LIFECYCLE MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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(240701329)

in partial fulfilment of the End Semester Examination for the course

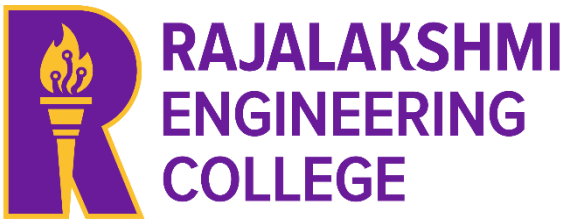
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

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BONAFIDE CERTIFICATE

Certified that this project report titled "ASSET LIFECYCLE MANAGEMENT SYSTEM" is the Bonafide work of MOHIDEEN ABDUL KADER ML (240701329), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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DECLARATION

I, MOHIDEEN ABDUL KADER ML (240701329) , hereby declare that the project report titled “ASSET LIFECYCLE MANAGEMENT SYSTEM”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

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I also acknowledge Zoho Corporation for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

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ABSTRACT.

Managing the ownership, condition, maintenance, warranty, and lifecycle history of physical assets is a significant challenge for organisations relying on manual records and spreadsheets. These approaches often result in fragmented information, data inconsistencies, delayed maintenance, difficulty tracking ownership changes, and missed warranty deadlines. These limitations affect administrators, asset managers, technicians, and asset owners who require accurate information throughout an object's lifecycle. To address these challenges, the Asset Lifecycle Management System was developed as a centralised platform for managing physical assets.

The application was developed using Zoho Creator, a low-code platform that enables rapid development through visual components, forms, workflows, and custom scripting. The system comprises nine interconnected forms covering object registration, owner management, warranty management, inspections, ownership transfers, maintenance requests, service providers, and service records. Reports and four custom pages provide access to operational information and administrative dashboards. Deluge scripting automates lifecycle history generation, ownership updates, maintenance completion, and warranty expiry reminders. A Blueprint-based approval process manages ownership transfers through defined stages. Role definitions for administrators, object managers, inspectors, and approvers support controlled access to application functions.

The application was functionally tested using sample records and scenarios involving object registration, ownership transfers, inspections, maintenance completion, and warranty reminders. Workflow execution, automatic record creation, data relationships, and report accessibility were verified through the live application. The application was deployed using Zoho Creator's deployment environment. By consolidating nine forms and automating key operational activities, the system reduces repetitive manual updates and improves information accessibility, traceability, and consistency. Although processing-time improvements have not been quantitatively benchmarked, the application provides a foundation for more efficient asset lifecycle management.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Role-Based Access Control; Asset lifecycle management system.

PROJECT AT A GLANCE

Attribute	Details
Project Title	ASSET LIFECYCLE MANAGEMENT SYSTEM
Application Domain	Asset Management and Digital Product Lifecycle Management (DPLM).
Student Name & Register No.	Mohideen Abdul Kader ML, 240701329
Zoho Creator Edition	Free
Application Name (Link Name)	Zoho Creator
Published / Live URL	https://digitalobjectpassportsystem.zohocreatorportal.in/
Demo Credentials (per role)	Admin: mohideenabdulkader.ml.2024.cse@rajalakshmi.edu.in ||
Number of Forms / Subforms	9/2
Number of Reports	9
Number of Pages / Dashboards	4
Number of Workflows	Form : 4 / Schedule : 1 / Blueprint :1
Deluge Functions / Custom Actions	4
Integrations	Zoho Mail
Roles & Profiles Defined	Administrator, Object Manager, Inspector , Approver
Environments Used	Development → Stage → Production
Date of Production Deployment	10 November 2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
JSON	JavaScript Object Notation
LCAP	Low-Code Application Platform
OAuth	Open Authorization
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
SSO	Single Sign-On
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

The Asset Lifecycle Management System belongs to the domain of Asset Management and Digital Product Lifecycle Management. Asset management involves the systematic tracking, monitoring, maintenance, and management of physical assets throughout their useful life. Organisations own and manage various assets, including computers, laptops, office equipment, electronic devices, machinery, and other resources required for their daily operations. Maintaining accurate information about these assets is essential for ensuring operational efficiency, accountability, and effective resource utilisation.

Digital Product Lifecycle Management focuses on maintaining a continuous record of an object's identity, ownership, condition, maintenance activities, warranty information, and lifecycle events. A Asset lifecycle Management provides a centralised digital record containing these details, allowing authorised users to access relevant information throughout the object's lifecycle. It improves traceability by maintaining historical records of important events such as registration, inspection, repair, ownership transfer, and maintenance.

In a conventional asset management environment, information about physical assets is often maintained using spreadsheets, paper-based records, separate documents, and manual communication. When a new asset is purchased, an administrator records its details, including the asset name, category, purchase date, purchase price, and assigned owner. Supporting documents such as invoices, warranty certificates, and purchase receipts are stored separately.

When an asset requires inspection or maintenance, the concerned employee reports the issue to the responsible administrator or technician. The request is reviewed, and maintenance activities are assigned manually. Details about the problem, service provider, repair work, spare parts, and maintenance expenses are recorded separately. This makes it difficult to maintain a consolidated history of an asset's condition and maintenance activities.

Ownership transfers introduce another administrative process. When an asset is reassigned from one employee to another, the transfer request must be reviewed and approved by the appropriate authority. The previous ownership details and updated

assignment must be maintained accurately to ensure accountability. Similarly, warranty expiry dates must be monitored manually, which can result in missed deadlines and delays in servicing or replacement claims.

As the number of assets increases, these processes become more difficult to coordinate. Maintaining information across multiple spreadsheets and documents can lead to duplicate entries, inconsistent records, delayed updates, and difficulty retrieving historical information. Administrators may also spend considerable time checking warranty dates, following up on maintenance requests, verifying ownership details, and updating asset records.

The proposed Asset Lifecycle Management System addresses these requirements by providing a centralised platform for managing physical assets and their associated records. It aims to improve information accessibility, maintain lifecycle traceability, support controlled ownership transfers, and simplify maintenance and warranty management. By replacing fragmented manual processes with an integrated digital solution, the application establishes a more organised and efficient approach to physical asset lifecycle management.

1.2 Problem Statement

The organisation currently manages its physical assets through spreadsheets, paper-based documents, and separate manual processes for registration, ownership tracking, warranty management, inspections, and maintenance. The absence of a centralised system makes it difficult to maintain accurate and consistent asset information throughout its lifecycle. Administrators, asset managers, technicians, and approvers face challenges in coordinating activities, retrieving historical records, monitoring warranties, and tracking ownership changes. Consequently, these processes increase administrative effort, create opportunities for errors, delay decision-making, and reduce overall visibility into asset management operations.

Key Pain Points

- **Fragmented Data Management:** Asset information, ownership details, warranty certificates, and maintenance records are stored across multiple spreadsheets and documents, resulting in data duplication and inconsistencies.

- **Manual Maintenance Tracking:** Inspection schedules, maintenance requests, repair details, and service expenses require manual recording and follow-up, increasing administrative effort.
- **Delayed Ownership Approvals:** Ownership transfers involve manual verification and approval, making it difficult to track request status and maintain updated ownership records.
- **Lack of Lifecycle Visibility:** There is no unified view of an object's registration, condition, warranty, maintenance, and ownership history, making information retrieval difficult.
- **Limited Automation and Access Control:** Manual warranty monitoring and record updates increase the possibility of missed deadlines and errors, while inadequate role-based access restrictions can expose information to unauthorised modifications.

1.3 Objectives

The primary objectives of the Asset Lifecycle Management System are:

- **To design** a structured data model and interconnected forms for managing object registration, ownership, warranty information, inspections, maintenance activities, and lifecycle records.
- **To automate** object registration history, ownership updates, maintenance completion, and warranty expiry reminders using Deluge scripting and Zoho Creator workflows.
- **To implement** a Blueprint-based approval process for managing ownership transfers through defined stages and ensuring controlled record updates.
- **To develop** role-based access and customised dashboards and reports for administrators, object managers, inspectors, and approvers to improve information accessibility and operational monitoring.
- **To establish** automated notifications and lifecycle tracking to improve maintenance coordination, warranty monitoring, and asset traceability.

- **To deploy** the application through Zoho Creator with appropriate user permissions, access controls, and data management practices to support secure and reliable asset management.

1.4 Scope of the Project

In-Scope

The project focuses on developing a web-based Asset Lifecycle Management System using Zoho Creator. Its scope includes:

- **Object Management:** Register objects and maintain their identification, ownership, and purchase details.
- **Warranty Management:** Store warranty documents, expiry dates, and related information.
- **Inspection and Maintenance:** Record inspections, condition assessments, maintenance requests, and service activities.
- **Ownership Transfers:** Manage ownership changes through a Blueprint-based approval process.
- **Lifecycle Tracking:** Maintain object history through automated lifecycle records.
- **Workflow Automation:** Use Deluge scripting for registration history, ownership updates, maintenance completion, and warranty reminders.
- **Dashboards and Reports:** Provide administrative dashboards and reports for monitoring asset information and operations.
- **User Management:** Support four roles: Administrator, Object Manager, Inspector/Technician, and Approver, with role-based permissions.

The application is primarily accessible through a web browser, with mobile access subject to Zoho Creator's supported features and configuration.

1.5 Justification for a Low-Code Approach

Zoho Creator was selected for developing the Asset Lifecycle Management System because it provides a low-code environment that simplifies application development

and maintenance. Its visual development tools enable rapid creation of forms, reports, dashboards, and workflows with minimal programming effort.

The platform offers several advantages:

Development Speed: Drag-and-drop tools accelerate application development.

Easy Maintenance: Visual configurations and Deluge scripting simplify modifications.

Built-in Hosting: Cloud hosting eliminates independent server management.

Mobile Accessibility: Applications can be accessed through web browsers and mobile devices.

Security: Role-based permissions and authentication help protect organisational data.

Cost Effectiveness: Reduced coding and infrastructure requirements lower development and maintenance effort.

Additionally, Zoho Creator supports Deluge scripting, workflow automation, Blueprint processes, and external integrations, making it suitable for managing object registration, ownership transfers, maintenance, and warranty activities.

Reference: Zoho Corporation, *Zoho Creator – Low-Code Application Development Platform*.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The existing asset management process relies on spreadsheets, paper-based documents, and manual communication between administrators, asset owners, technicians, and approvers. The process involves the following steps:

1. Asset Registration: Administrators manually record newly purchased assets, including their names, categories, purchase details, and assigned owners.

2. Document Management: Invoices, warranty certificates, and supporting documents are stored separately in physical files or digital folders.

3. Ownership Management: Asset ownership changes are handled through manual requests, approvals, and spreadsheet updates.

4. Inspection and Maintenance: Owners report asset issues, technicians inspect the assets, and maintenance activities are recorded manually.

5. Warranty Monitoring: Administrators periodically check warranty expiry dates and notify owners about upcoming expirations.

6. Record Maintenance: Inspection results, service expenses, ownership changes, and lifecycle events are maintained in separate records.

7. Report Generation: Administrators manually collect and consolidate information from different sources to prepare asset management reports.

The existing process is time-consuming and fragmented, making it difficult to maintain accurate records, monitor asset conditions, track ownership changes, and retrieve complete lifecycle information efficiently.

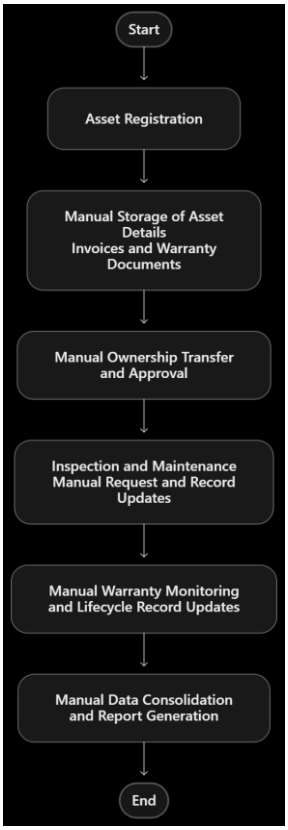


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- Fragmented Data Management: Asset details, ownership records, warranty documents, and maintenance information are stored separately, making data retrieval difficult.

- Manual Process and Delays: Ownership approvals, maintenance updates, and warranty monitoring depend on manual intervention, increasing processing time.

- Lack of Lifecycle Visibility: The absence of a centralised system makes it difficult to track an asset's complete history, current condition, ownership, and maintenance activities.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	8–10 weeks.	2–3 weeks.

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Skills required	Programming, frontend, backend, database	Basic coding, Deluge, visual tools
Hosting & infrastructure	Manual server and database setup	Built-in cloud hosting and database infrastructure.
Mobile application	Separate development required	Built-in Mobile access
Security & access control	Custom implementation	Built-in roles and permissions
Maintenance & change	Code modifications and redeployment	Visual configuration and Deluge
Customisation limits	High flexibility	Platform-dependent flexibility
Cost	Higher development and maintenance cost.	Lower initial development cost

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Bucaioni et al. (2022)	Modelling in Low-Code Development: A Multi-Vocal Systematic Review	Low-code development uses visual modelling and abstraction to simplify software development.	Supports choosing a low-code platform for application development.
2	Pinho et al. (2023)	What About the Usability in Low-Code Platforms?	Highlights usability as an important factor in low-code platform adoption.	Supports the use of visual forms and user-friendly interfaces.
3	Zoho Corporation	Zoho Creator – Workflows and Blueprints Documentation	Provides workflow automation, Deluge scripting, and Blueprint-based process management.	Supports automated lifecycle updates and ownership approval workflows

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
5	IBM	Maximo Application Suite – Asset Management	Provides asset tracking, maintenance management, inspections,.	Serves as a reference for the asset management features included in this project.

2.5 Emerging Trends in Low-Code Development

1. AI-Assisted App Building: AI tools help generate application components, suggest workflows, and simplify coding, making application development faster.

2. Citizen Development: Low-code platforms enable users with limited programming knowledge to build applications using visual tools. This influenced the use of Zoho Creator's drag-and-drop forms, reports, and pages.

3. Hyper-Automation: It focuses on automating repetitive business processes using workflows and scripts. This influenced the use of Deluge scripts for lifecycle updates, ownership transfers, maintenance activities, and warranty reminders.

4. Composable Applications: Applications are developed by combining reusable components. This influenced the modular design of the system using separate forms, reports, workflows, and dashboards for different asset management activities.

Overall: These trends encouraged a modular and low-code approach, reducing manual work while making the application easier to develop, maintain, and extend.

2.6 Proposed System

The proposed Asset Lifecyle Management System is developed using Zoho Creator to centralise asset information and automate asset management activities. It includes forms, reports, dashboards, workflows, and Blueprint-based ownership approvals.

The system overcomes the limitations identified in Section 2.2 as follows:

1. Fragmented Data Management: Centralises asset details, ownership records, warranty documents, and maintenance information in one application.

2. Manual Process and Delays: Automates ownership updates, lifecycle history creation, maintenance updates, and warranty reminders using Deluge scripts and workflows.

3. Lack of Lifecycle Visibility: Maintains asset history, ownership details, inspection records, and maintenance activities through centralised reports and dashboards.

Overall: The proposed system improves data organisation, reduces manual work, and provides better visibility into the complete asset lifecycle.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Has complete control over the application.	Manage users, configure settings, access all records.
Object Manager	Manages asset information and monitors asset activities.	Register objects, update records, monitor dashboards, oversee ownership transfers.
Inspector / Technician	Handles asset inspections and maintenance activities.	Conduct inspections, update object conditions, manage service records.
Approver	Reviews and manages ownership transfer approvals.	Reviews and manages ownership transfer approvals.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow users to register objects by entering mandatory details such as object name, ID, category, purchase date, and owner.	Administrator, Object Manager	High
FR-02	The system shall automatically calculate the age of an object based on its purchase date.	Administrator, Object Manager	High
FR-03	The system shall automatically generate lifecycle history records when an object is registered or its ownership is transferred.	Administrator, Object Manager	High
FR-04	The system shall manage ownership transfer requests through a Blueprint approval process with Requested, Under Review, Approved, Rejected, and Completed stages.	Object Manager, Approver	High
FR-05	The system shall provide dashboards and reports to view object details, ownership information, maintenance records, and lifecycle history.	Administrator, Object Manager	High
FR-06	The system shall trigger warranty expiry reminders through a scheduled workflow for eligible records.	Administrator, Object Manager	Medium

FR ID	Requirement	Role(s)	Priority
FR-07	The system shall update object maintenance information when a service record is marked as completed.	Administrator, Inspector / Technician	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports containing up to 500 records should load within 5 seconds under normal conditions.
NFR-02	Security	Users shall access only the data and functions permitted by their assigned roles.
NFR-03	Usability	The application shall be accessible through web browsers and the Zoho Creator mobile application.
NFR-04	Availability	The application shall be accessible whenever the Zoho Creator cloud service is available.
NFR-05	Maintainability	Forms, workflows, and reports shall be easy to modify and maintain using Zoho Creator.
NFR-06	Auditability	Important asset lifecycle events, including registration and ownership transfers, shall be recorded for future reference.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator Education Edition
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing tool	Postman
Other Zoho services	Zoho Mail,
Mobile	Zoho Creator mobile app (Android / iOS)
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.4 Usecase of the Application :

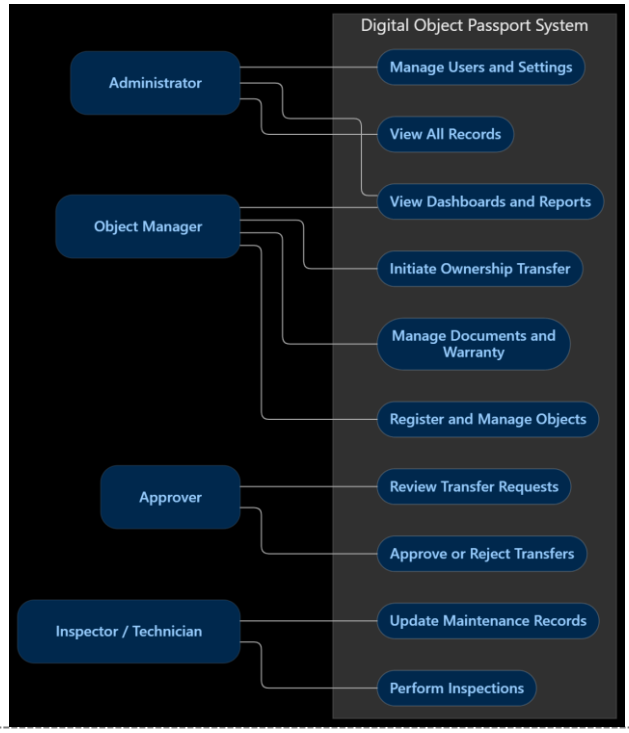


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Object Registration

Use Case ID	UC-01
Use Case Name	Object Registration
Actor(s)	Administrator, Object Manager
Pre-condition	The user must be logged in and have permission to register objects
Main Flow	1. The user opens the Object Registration form. 2. The user enters the required object details. 3. The user submits the form. 4. The system validates the mandatory fields. 5. The system saves the object record and calculates its age. 6. The system automatically creates a lifecycle history entry.

Alternate Flow	If mandatory fields are missing or invalid, the system displays an error and requests corrections.
Post-condition	The object record is successfully stored, and its registration event is added to the lifecycle history.
Related FR	FR-01, FR-02, FR-03

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

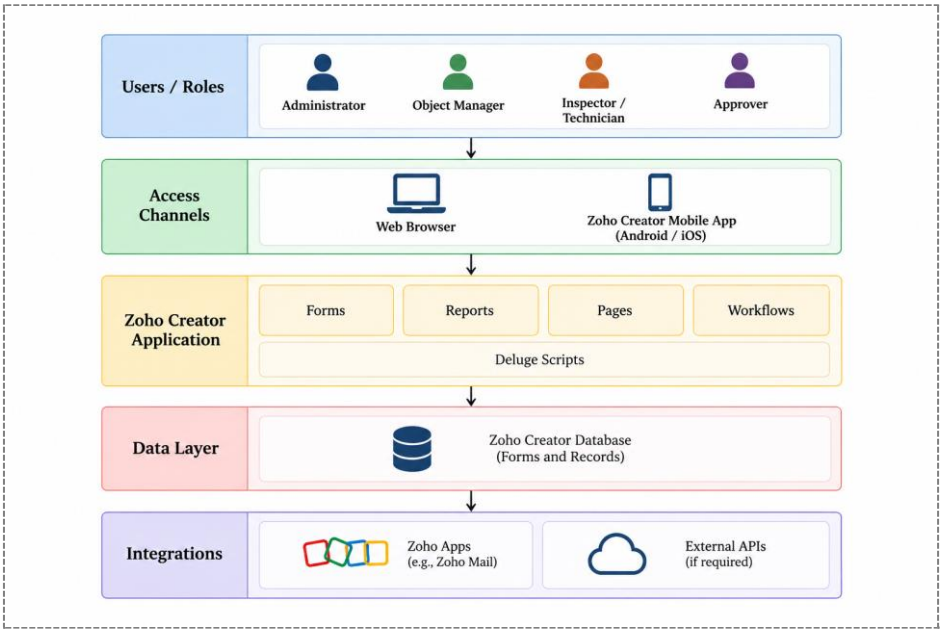


Figure 4.1: System Architecture of the Asset Lifecycle Management System

1. User and Access Layer: Includes four roles: Administrator, Object Manager, Inspector / Technician, and Approver. Users access the application through a web browser or the Zoho Creator mobile application.

2. Application Layer: Built using Zoho Creator. It provides forms for data entry, reports for viewing records, and pages for dashboards and application navigation.

3. Business Logic Layer: Contains Deluge scripts, workflows, and Blueprint. It handles automatic lifecycle updates, object age calculation, maintenance updates, warranty reminders, and ownership approval processes.

4. Data Layer: Stores object details, owner information, warranty documents, inspection records, maintenance details, ownership transfers, and lifecycle history using Zoho Creator forms and its underlying database.

5. Integration Layer: Represents communication with Zoho services and external APIs through connections. The current implementation primarily uses Zoho Creator and does not include confirmed external API integrations.

Overall: The layered architecture separates user interaction, application functionality, business logic, and data storage, making the system easier to manage and maintain.

4.2 Data Model

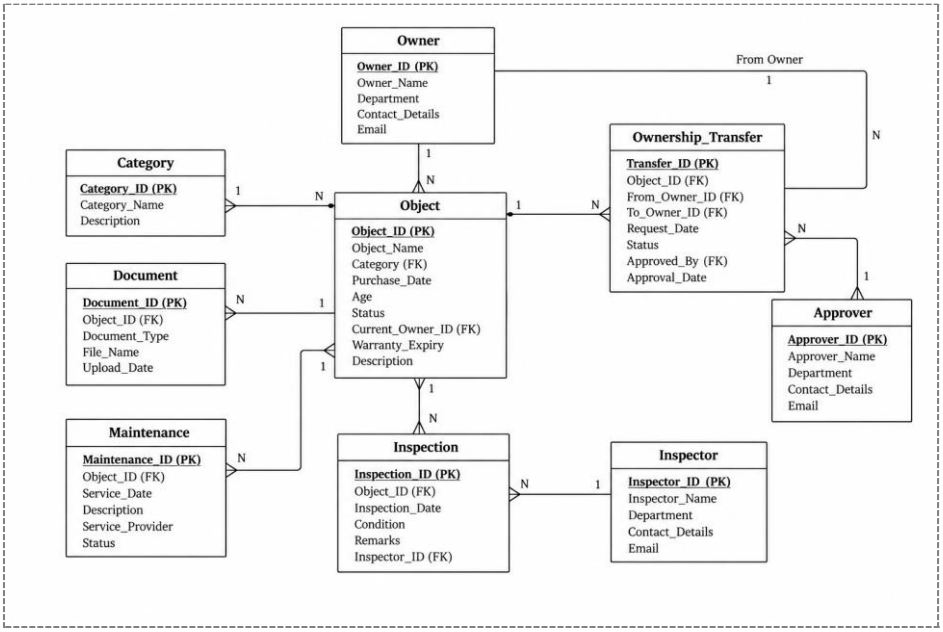


Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Objects	Object ID, Object Name, Purchase Date, Current Condition	Owners	Many-to-One	Lookup in Objects
Owners	Owner ID, Owner Name, Contact Details, Email	Objects	One-to-Many	Lookup in Objects
Documents & Warranty	Document ID, Document Type, Warranty Expiry	Objects	Many-to-One	Lookup in Documents & Warranty
Inspections & Maintenance	Inspection ID, Inspection Date, Condition	Objects	Many-to-One	Lookup in Inspections & Maintenance
Maintenance Requests	Request ID, Request Date, Status	Objects	Many-to-One	Lookup in Maintenance Requests

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Ownership Transfers	Transfer ID, Transfer Date, Transfer Status	Objects, Owners	Many-to-One	Lookup fields

4.3 Form Design

Table 4.2: Field Specification – Objects Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Object ID	Object_ID	Auto Number	No	System generated	Unique object identifier
Object Name	Object_Name	Single Line	Yes	Cannot be empty	Name of the object
Category	Category	Drop Down	Yes	Select from available categories	Object classification
Purchase Date	Purchase_Date	Date	Yes	Valid date	Date of purchase
Owner	Owners	Lookup	Yes	Select from Owners form	Current object owner
Object Age	Object_Age	Formula	No	Automatically calculated using Purchase Date	Age of the object in years
Current Condition	Current_Condition	Drop Down	No	Select from configured options	Current physical condition
Description	Description	Multi Line	No	—	Additional object details
QR Code	Object_QR_Code	Verify in Creator	No	Based on Object ID	Object identification

Table 4.2: Field Specification – Owners Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Owner ID	Owner_ID	Auto Number	No	System generated	Unique owner identifier
Owner Name	Owner_Name	Single Line	Yes	Cannot be empty	Name of the owner
Email	Email	Email	Yes	Valid email format	Owner's email address
Phone Number	Phone_Number	Phone	No	Valid phone number	Contact information
Address	Address	Address	No	—	Owner's address
Department	Department	Single Line	No	—	Owner's department

Table 4.2: Field Specification – Documents & Warranty Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Document ID	Document_ID	Auto Number	No	System generated	Unique document identifier
Object	Object	Lookup	Yes	Select from Objects form	Associated object
Document Type	Document_Type	Drop Down	Yes	Select document category	Type of document
Document Name	Document_Name	Single Line	Yes	Cannot be empty	Document title
Document File	Document_File	File Upload	Yes	Supported file formats	Uploaded document

Warranty Start Date	Warranty_Start_Date	Date	No	Valid date	Warranty commencement date
Warranty Expiry Date	Warranty_Expiry_Date	Date	No	Must be later than start date	Warranty expiration date
Remarks	Remarks	Multi Line	No	—	Additional information

Table 4.2: Field Specification – Inspection and Maintainence Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Inspection ID	Inspection_ID	Auto Number	No	System generated	Unique inspection identifier
Object	Object	Lookup	Yes	Select from Objects form	Inspected object
Inspection Date	Inspection_Date	Date	Yes	Valid date	Date of inspection
Inspector	Inspector	Single Line	Yes	Cannot be empty	Name of the inspector
Condition	Condition	Drop Down	Yes	Select configured condition	Observed object condition
Remarks	Remarks	Multi Line	No	—	Inspection observations
Maintenance Required	Maintenance_Required	Decision Box	No	Default: No	Indicates whether maintenance is required

Table 4.2: Field Specification – Maintenance Requests Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Request ID	Request_ID	Auto Number	No	System generated	Unique request identifier
Object	Object	Lookup	Yes	Select from Objects form	Object requiring maintenance
Request Date	Request_Date	Date	Yes	Default: Current Date	Date of request
Requested By	Requested_By	Single Line	Yes	Cannot be empty	Name of the requester
Issue Description	Issue_Description	Multi Line	Yes	Cannot be empty	Details of the reported issue
Priority	Priority	Drop Down	Yes	Low / Medium / High	Maintenance priority
Request Status	Request_Status	Drop Down	Yes	Default: Pending	Current request status

Table 4.2: Field Specification – Ownership Transfers Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Transfer ID	Transfer_ID	Auto Number	No	System generated	Unique transfer identifier
Object	Object	Lookup	Yes	Select from Objects form	Object being transferred
Current Owner	Current_Owner	Lookup	Yes	Select from Owners form	Existing owner

New Owner	New__Owner	Lookup	Yes	Select from Owners form	Proposed new owner
Transfer Date	Transfer_Date	Date	Yes	Valid date	Date of transfer request
Transfer Status	Transfer_Status	Drop Down	Yes	Default: Requested	Current transfer status
Remarks	Remarks	Multi Line	No	—	Additional transfer details

Table 4.2: Field Specification – Ownership Transfers Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
History ID	ID	Auto Number	No	System generated	Unique history identifier
Object	Object	Lookup	Yes	Select from Objects form	Associated object
Event Type	Event_Type	Drop Down	Yes	Select event category	Type of lifecycle event
Event Date	Event_Date	Date	Yes	Default: Current Date	Date of event
Event Description	Event_Descr	Multi Line	No	—	Details of the lifecycle event

Table 4.2: Field Specification – Service Providers Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Provider ID	Provider_ID	Auto Number	No	System generated	Unique provider identifier

Provider Name	Provider_Name	Single Line	Yes	Cannot be empty	Name of the service provider
Contact Person	Contact_Person	Single Line	No	—	Provider representative
Email	Email	Email	Yes	Valid email format	Provider email address
Phone Number	Phone_Number	Phone	No	Valid phone number	Contact number
Address	Address	Address	No	—	Provider address
Service Type	Service_Type	Drop Down	Yes	Select configured service type	Services offered

Table 4.2: Field Specification – Service Records Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Service Record ID	Service_Record_ID	Auto Number	No	System generated	Unique service record identifier
Object	Object	Lookup	Yes	Select from Objects form	Serviced object
Service Provider	Service_Provider	Lookup	Yes	Select from Service Providers form	Provider handling the service
Service Date	Service_Date	Date	Yes	Valid date	Date of service
Service Description	Service_Description	Multi Line	Yes	Cannot be empty	Details of work performed
Service Cost	Service_Cost	Currency	No	Must be zero or greater	Total service cost
Service Status	Service_Status	Drop Down	Yes	Default: Pending	Current service status

Remarks	Remarks	Multi Line	No	—	Additional service information

4.4 Process and Workflow Design

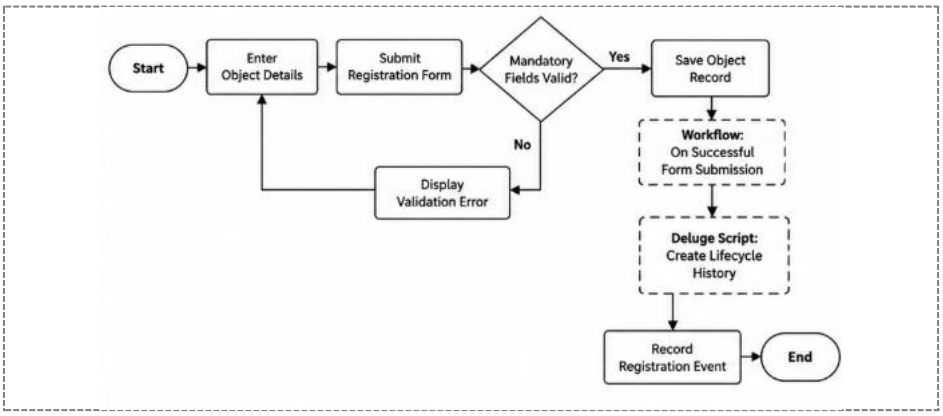


Figure 4.3: Workflow / Process Flow for Object Registration

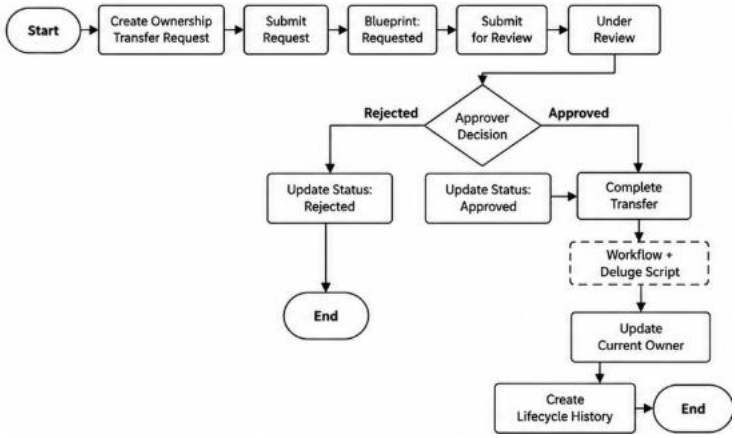


Figure 4.4: Workflow / Process Flow for Ownership Transfer and Approval

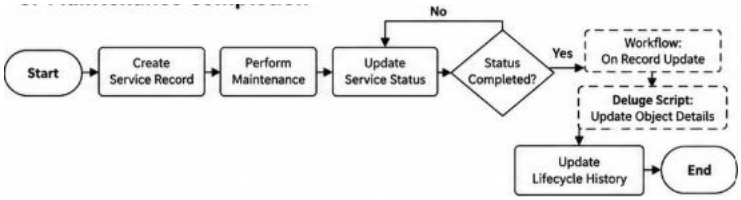


Figure 4.5 Workflow / Process Flow for Maintenance Completion

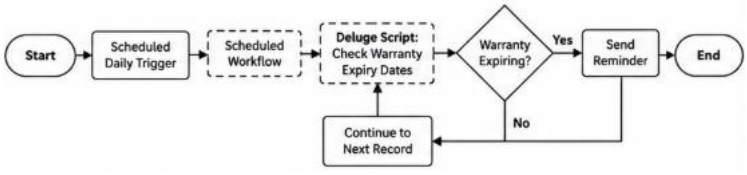


Figure 4.6: Workflow/ Process Flow for Warranty Expiry Remainder

4.5 User Interface and Navigation Design

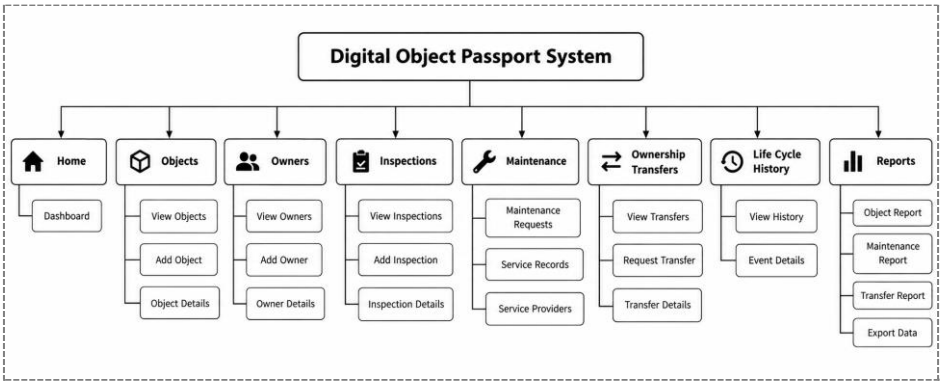


Figure 4.4: Navigation Structure of the System

4.6 Access Control Design

Table 4.3: Role–Permission Matrix

Component	Administrator	Object Manager	Inspector / Technician	Approver
Objects Form	A E V D	A E V	V	V
Owners Form	A E V D	A E V	V	V
Documents & Warranty Form	A E V D	A E V	V	V
Inspections & Maintenance Form	A E V D	V	A E V	V
Maintenance Requests Form	A E V D	A E V	A E V	V
Ownership Transfers Form	A E V D	A E V	—	A E V
Lifecycle History Form	A E V D	V	V	V
Service Providers Form	A E V D	A E V	V	—
Service Records Form	A E V D	V	A E V	V

CHAPTER 5
IMPLEMENTATION.

5.1 Application Setup

The Asset Lifecycle Management System was developed using Zoho Creator, a low-code application development platform. The application was created from scratch using the Zoho Creator workspace without using a predefined template.

The application was designed to manage object registration, ownership, documentation, inspections, maintenance, and lifecycle history in a centralised environment.

The setup involved the following steps:

1. Workspace Creation: A new application was created in the Zoho Creator workspace.

2. Application Development: Forms, reports, pages, and workflows were created to support different asset management activities.

3. Navigation Structure: The application was organised into sections such as Objects, Owners, Inspections, Maintenance, Ownership Transfers, Lifecycle History, and Reports.

4. Theme and Branding: A consistent interface was maintained across the application for easy navigation and usability.

5. Access Setup: Four user roles were defined: Administrator, Object Manager, Inspector / Technician, and Approver.

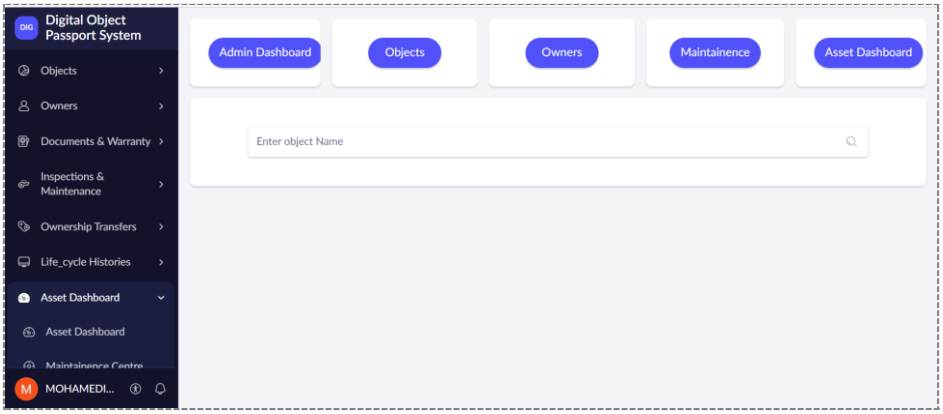


Figure 5.1: Home Page

5.2 Forms and Data Capture

The Asset Lifecycle Management System consists of nine forms developed using Zoho Creator. These forms organise asset-related information and establish relationships between records using lookup fields.

5.2.1 Objects Form

The Objects form is the central form of the application. It stores object identification, category, purchase date, current owner, condition, and description.

- Field Properties: Auto Number for Object ID, Single Line for object name, Date for purchase date, Lookup for owner, and Formula for object age.
- Lookups: The Owner field connects to the Owners form.
- Validation: Mandatory fields are configured to ensure essential object information is entered.
- Automation: Object age is calculated automatically using the purchase date. A workflow creates a lifecycle history entry after successful registration.

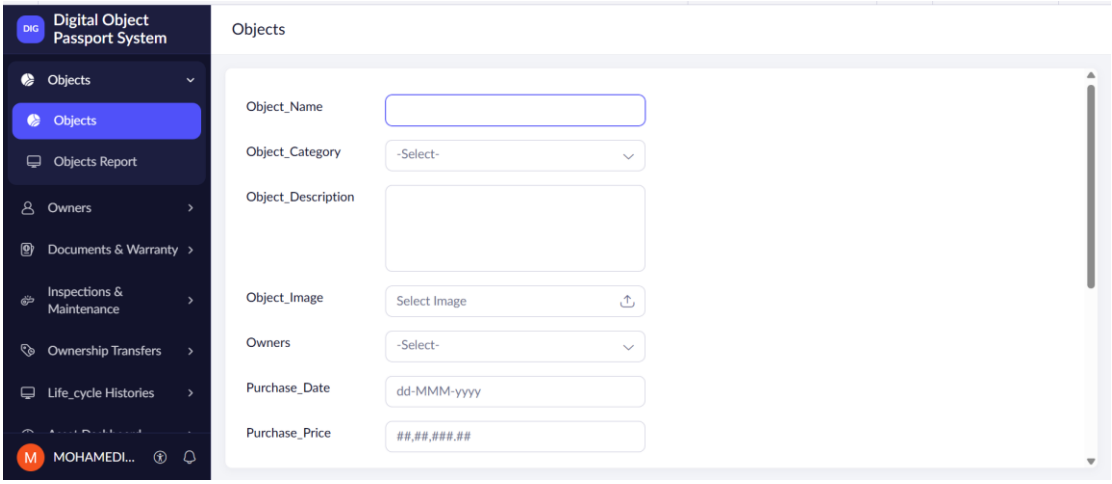


Figure :5.2 Objects Forms

5.2.2 Owners Form

Stores owner information such as name, email, phone number, and address.

- Field Properties: Single Line, Email, Phone, and Address fields.
- Validation: Email fields use email-format validation.
- Relationship: Owner records are linked to objects through a Lookup field.

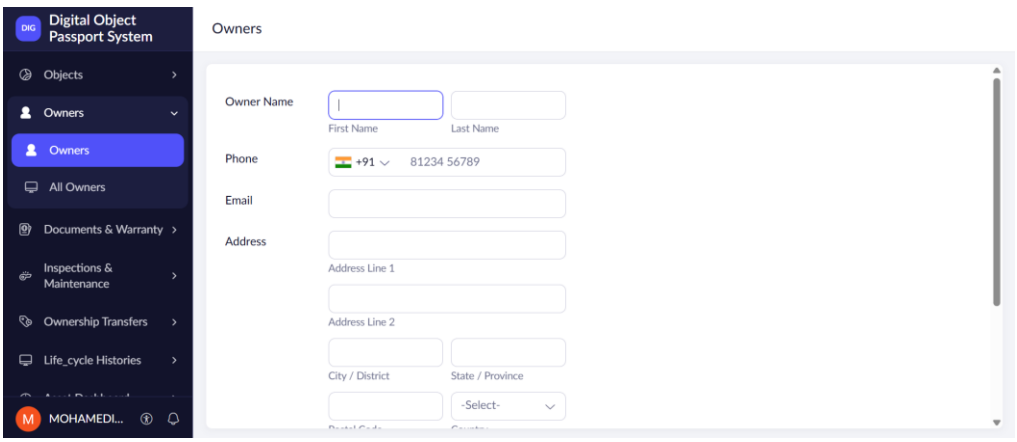


Figure: 5.3 Owners Form

5.2.3 Documents & Warranty Form

Stores documents and warranty information associated with registered objects.

- Field Properties: Lookup, Drop Down, Date, and File Upload fields.
- Lookups: Connects documents to the corresponding object.
- Validation: File and date fields are configured according to their input requirements.

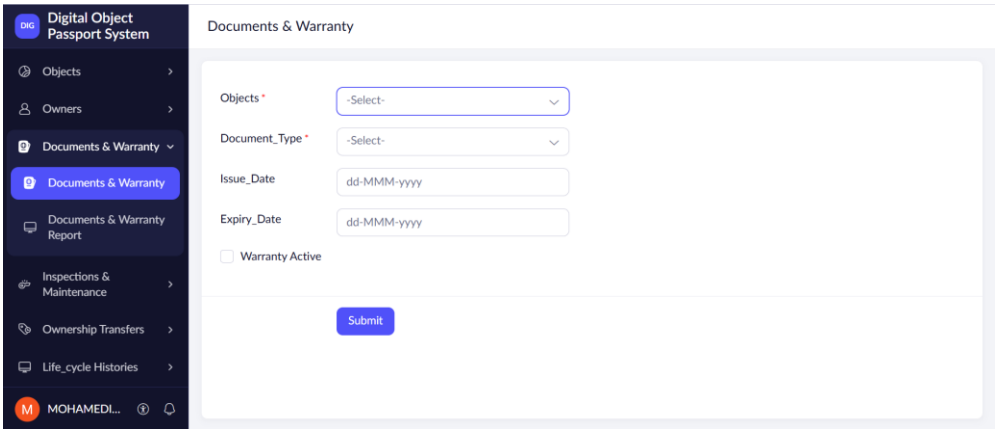


Figure: 5.4 Documents & Warranty Form

5.2.4 Inspections & Maintenance Form

Maintains inspection details and records the condition of objects.

- Field Properties: Lookup, Date, Drop Down, and Multi Line fields.
- Lookups: Links inspection records to the Objects form.
- Validation: Required inspection information is entered before submission.

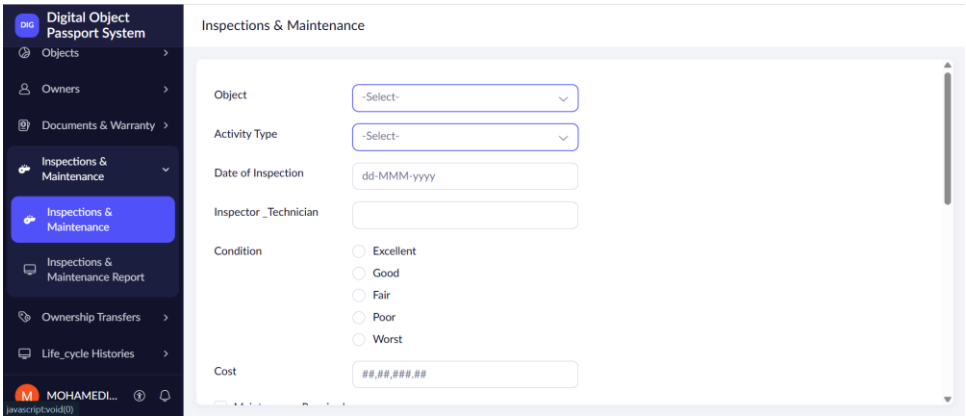


Figure: 5.5 Inspections and Maintenance Form

5.2.5 Maintenance Requests Form

Allows users to record maintenance requests for registered objects.

- Field Properties: Lookup, Date, Drop Down, and Multi Line fields.
- Lookups: Connects each request to its corresponding object.
- Validation: Required request details are provided before submission.

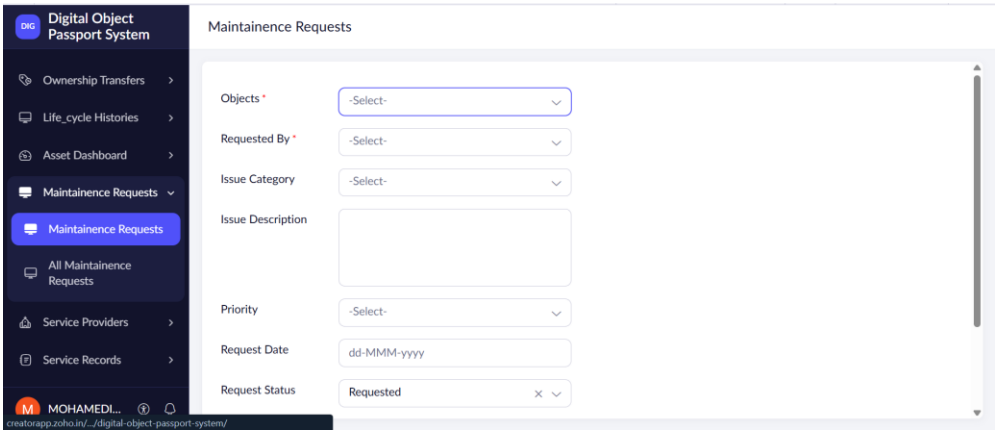


Figure: 5.6 Maintenance Requests

5.2.6 Ownership Transfers Form

Stores ownership transfer requests and their approval status.

- Field Properties: Lookup, Date, and Drop Down fields.
- Lookups: Connects the transfer request to the object and new owner.

- Validation: Object and new owner details are required.
- Automation: Ownership transfers are managed through a Blueprint approval process.

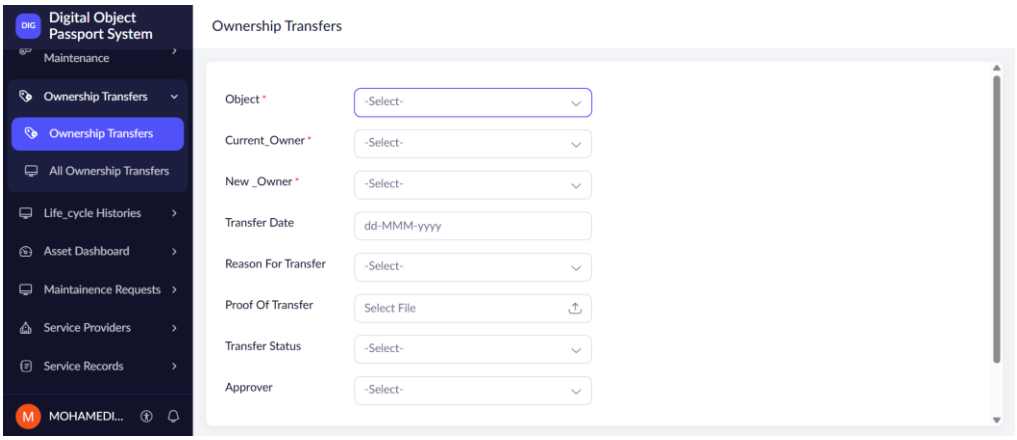


Figure: 5.7 Ownership Transfers

5.2.7 Life_Cycle History Form

Maintains a chronological record of important events associated with each object.

- Field Properties: Lookup, Drop Down, Date, and Multi Line fields.
- Lookups: Connects each history entry to an object.
- Automation: Lifecycle records are created automatically during object registration and ownership transfer completion.

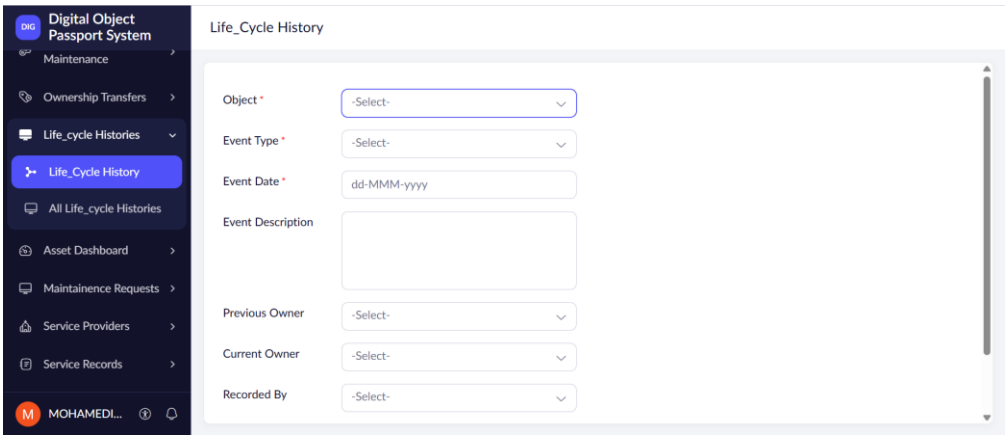


Figure: 5.8 Life_Cycle History Form

5.2.8 Service Providers Form

Stores information about service providers responsible for maintenance activities.

- Field Properties: Single Line, Email, Phone, and Address fields.
- Validation: Contact information is entered using the appropriate field formats.

- Relationship: Service providers are associated with service records through a Lookup field.

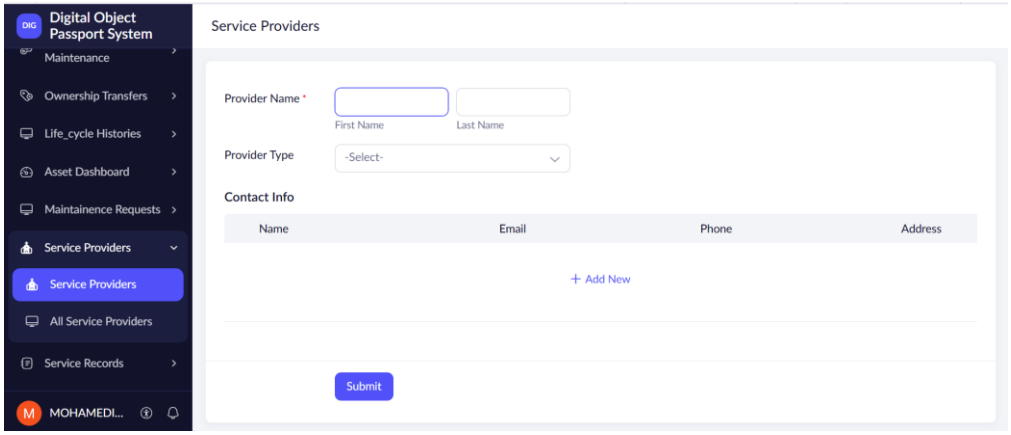


Figure: 5.9 Service Providers

5.2.9 Service Records Form

Maintains records of maintenance and servicing activities performed on objects.

- Field Properties: Lookup, Date, Currency, Drop Down, and Multi Line fields.
- Lookups: Connects service records to Objects and Service Providers.
- Automation: A workflow handles service completion and the associated object and lifecycle updates.

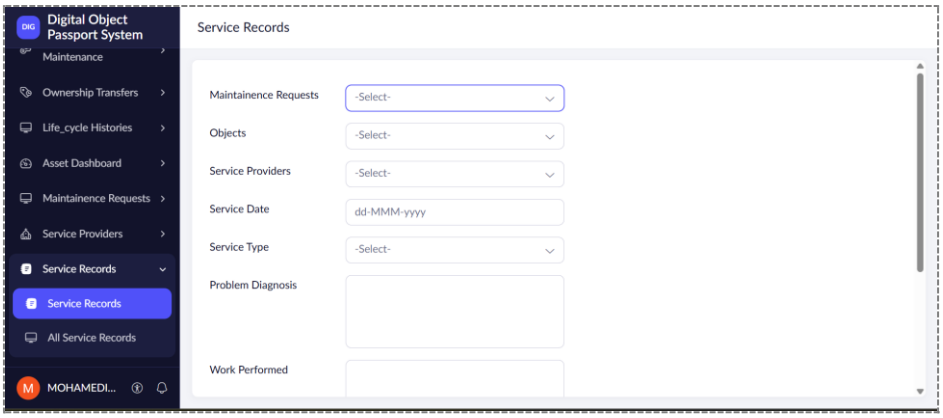


Figure 5.10: Service Records Form

5.3 Reports

Zoho Creator provides reports to view, filter, and manage records stored in the application. Nine reports were created corresponding to the application's forms. The reports help administrators and other users monitor asset information, ownership, inspections, and maintenance activities.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Objects Report	List	Objects	Filter by category and condition	View / Edit	Administrator, Object Manager
All Owners	List	Owners	Sort by owner name	View / Edit	Administrator, Object Manager
Documents & Warranty Report	List	Documents & Warranty	Filter by object and warranty expiry	View / Edit	Administrator, Object Manager
Inspections & Maintenance Report	List	Inspections & Maintenance	Sort by inspection date	View / Edit	Administrator, Inspector / Technician
All Maintenance Requests	List	Maintenance Requests	Filter by request status and priority	View / Edit	Administrator, Object Manager, Inspector / Technician
All Ownership Transfers	List	Ownership Transfers	Filter by transfer status	View / Edit	Administrator, Object Manager, Approver
All Life_Cycle Histories	List	Life_Cycle History	Sort by event date	View	Administrator, Object Manager
All Service Providers	List	Service Providers	Sort by provider name	View / Edit	Administrator, Object Manager
All Service Records	List	Service Records	Filter by service status and object	View / Edit	Administrator, Inspector / Technician

5.4 Pages and Dashboards

The Asset Lifecycle Management System contains four pages developed using Zoho Creator's Page Builder. Each page serves a specific purpose and provides users with easy access to important application features.

Home Page:

- Purpose: Acts as the main entry point for the application.
- Target Role: All users.
- Components Used: Navigation links, page headings, and shortcuts to important application sections.

Admin Dashboard

- Purpose: Provides administrators with an overview of asset management activities.
- Target Role: Administrator.
- Components Used: Summary panels, report links, and navigation elements for accessing application records.

Asset Dashboard

- Purpose: Provides an overview of registered objects and their current information.
- Target Role: Administrator and Object Manager.
- Components Used: Asset information panels, summary displays, and links to object-related reports.

Maintenance Centre

- Purpose: Provides a central interface for accessing maintenance-related activities.
- Target Role: Administrator, Object Manager, and Inspector / Technician.
- Components Used: Maintenance request links, service record access, and navigation elements.

Overall: The pages provide organised navigation and quick access to important asset management functions. They simplify interaction with the application and help users access information according to their responsibilities.

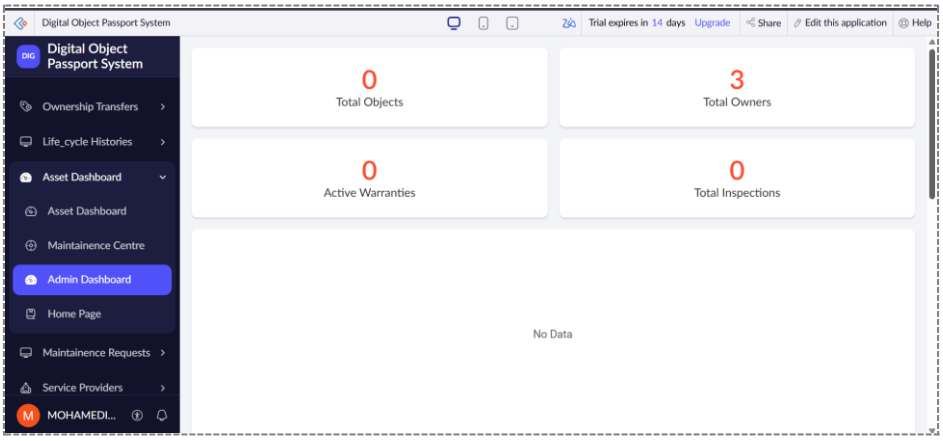


Figure 5.5: Admin Dashboard

5.5 Deluge Scripting

Deluge (Data Enriched Language for the Universal Grid Environment) is the scripting language used in Zoho Creator to implement application logic and automate business processes. In the Asset Lifecycle Management System, Deluge scripts are used to record object registration, update ownership details, maintain lifecycle history, and support maintenance and warranty-related workflows.

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Objects Form → On Success	Automatically create a lifecycle history entry when a new object is registered.	insert into, input, variable assignment, string concatenation
DS-02	Ownership Transfers → On Update	Update the object's current owner and record the ownership transfer when the transfer status becomes Completed.	if, record fetching, count(), record update, insert into
DS-03	Service Records → On Update	Handle the completion of a service record and update the related object and its maintenance history.	Workflow conditions, if-else, record fetching, record updates
DS-04	Scheduled Workflow → Daily	Check warranty records and support reminders for warranties approaching expiry.	Scheduled execution, criteria-based record fetching, date comparison

Code 5.1: Object Registration – Lifecycle History

Location: Objects Form → On Success

Purpose: Automatically create a lifecycle history entry after successful object registration.

```
// DS-01 | Objects Form | On Success
// Purpose: Record the registration event in lifecycle history

insert into Lifecycle_History
[
```

```
    Object = input.ID
    Event_Type = "Registered"
    Event_Date = zoho.currentdate
    Event_Descr = "Digital passport created for " + input.Object_Name
];
```

Logic:

1. The script executes after a new object record is successfully submitted.
2. It inserts a new record into the Lifecycle_History form.
3. The newly created object is linked to the history record.
4. The event type, current date, and registration description are stored.

Outcome: Every newly registered object automatically receives a corresponding lifecycle history entry.

Code 5.2: Ownership Transfer Completion

Location: Ownership Transfers Form → On Update

Purpose: Update the current owner and record the completed ownership transfer.

```
// DS-02 | Ownership Transfers | On Update
// Purpose: Update owner after transfer completion

if(input.Transfer_Status == "Completed")
{
    if(input.Object != null && input.New__Owner != null)
    {
        objectRecord = Objects[ID == input.Object];

        if(objectRecord.count() > 0)
        {
            objectRecord.Owners = input.New__Owner;

            insert into Lifecycle_History
            [
                Object = input.Object
                Event_Type = "Ownership Transfer"
                Event_Date = zoho.currentdate
                Event_Descr = "Ownership transferred to a new owner."
            ];
        }
    }
}
```

Logic:

1. The script checks whether the transfer status is Completed.
2. It verifies that an object and a new owner have been selected.
3. It fetches the corresponding object record.
4. The object's current owner is updated.
5. A new lifecycle history entry is created to record the transfer.

Outcome: The object's ownership information is updated automatically, and the transfer is recorded in its lifecycle history.

Explanation of the Remaining Workflows

DS-03 – Service Completion Update

When a service record is marked as Completed, the configured workflow handles the related object's maintenance information and lifecycle tracking. This reduces the need for manually updating the object's records after maintenance.

DS-04 – Warranty Expiry Reminder

A scheduled workflow checks warranty records for approaching expiry dates. It identifies relevant records and supports the warranty reminder process, helping administrators monitor warranty periods.

Result

The implemented Deluge scripts automate important operations in the Asset Lifecycle Management System. They reduce manual record updates, maintain lifecycle information, and improve consistency across object registration, ownership transfers, maintenance, and warranty management.

5.6 Workflows and Automation

Workflows and automation in Zoho Creator are used to execute predefined actions when specific events or conditions occur. In the Asset Lifecycle Management System, workflows automate lifecycle history creation, ownership updates, maintenance completion, and warranty monitoring. A Blueprint is also used to manage the ownership transfer approval process.

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Object Registration History	Form Workflow	Objects form → On Success	A new object is successfully created	Automatically create a lifecycle history record with registration details	DS-01
Complete Ownership Transfer	Form Workflow	Ownership Transfers form → On Update	Transfer status is Completed	Update the object's current owner and create a lifecycle history record	DS-02

Ownership Transfer Approval	Blueprint	Stage transitions	Transfer request moves through the defined approval stages	Manage transfer progression through Requested, Under Review, Approved, Rejected, and Completed stages	—
Service Completion Update	Form Workflow	Service Records form → On Update	Service status is Completed	Update the related object's maintenance information and lifecycle history	DS-03
Warranty Expiry Reminder	Schedule	Scheduled execution	Warranty records meet the configured expiry criteria	Check warranty expiry and support reminder notifications	DS-04
Maintenance Request Alert	Form Workflow	Maintenance request event	A maintenance request is submitted	Execute the configured maintenance request alert action	—

Note: The table lists the workflows identified in your application. The exact trigger and condition for the Maintenance Request Alert and Warranty Expiry Reminder should match their actual Zoho Creator configurations. DS-03 and DS-04 are references to workflow logic, not confirmation that the full code is identical to the descriptions in Section 5.5.

Blueprint – Ownership Transfer Approval

The Ownership Transfer Approval Blueprint controls the movement of an ownership transfer request through a defined sequence of stages.

Stages and transitions:

- Requested: A new ownership transfer request is created.
- Under Review: The request is submitted for approval.
- Approved: The approver accepts the transfer request.
- Rejected: The approver rejects the request.
- Completed: The approved transfer is completed, triggering the ownership update workflow.

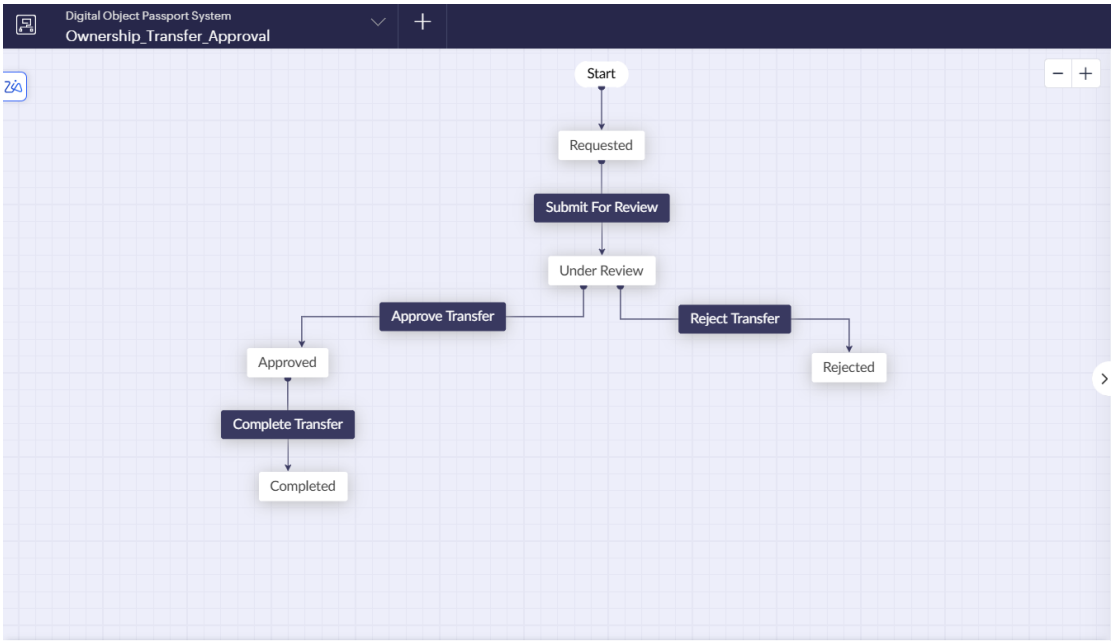


Figure: 5.6 Ownership Transfer Approval Blueprint

Result

The workflows automate repetitive operations and maintain consistency across the application. The Blueprint provides a controlled ownership approval process, while form and scheduled workflows support lifecycle tracking, maintenance updates, and warranty monitoring.

5.7 Integrations

Integration capabilities allow an application to exchange data with external services and other platforms. Zoho Creator supports several integration methods, including built-in integrations, connections, REST API calls using invokeurl, webhooks, and Creator REST APIs.

The Asset Lifecycle Management System primarily uses Zoho Creator's built-in capabilities, including forms, workflows, reports, and Deluge scripting. No external CRM, third-party REST API, or webhook integration has been confirmed as implemented in the application. Therefore, the documentation should distinguish between the platform's capabilities and integrations actually used.

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose

Zoho Creator Platform	Zoho Creator	Built-in platform functionality	Zoho account authentication	Manages application forms, reports, pages, workflows, and stored records
Deluge Workflow Automation	Zoho Creator Deluge environment	Built-in Deluge scripting and workflow execution	Handled by Zoho Creator	Creates lifecycle history records and updates ownership and maintenance information
QR Code Field	Zoho Creator QR Code field	Built-in field configuration	No separate API authentication	Generates a QR code using the object's Object ID as input data

Integration Method and Data Flow:

The application uses Zoho Creator as its primary platform. Its forms and workflows exchange data internally through Deluge scripts and lookup relationships.

For example, when an ownership transfer is completed:

1. The Ownership Transfers form provides the object and new owner details.
2. A Deluge workflow retrieves the corresponding object record.
3. The object's current owner is updated.
4. A lifecycle history record is inserted.

This is an internal application-level data exchange and does not require an external REST API call.

5.8 User Management and Role-Based Access

Role-Based Access Control (RBAC) is used to manage user access based on their responsibilities within the application. Zoho Creator provides user management and permission settings that allow administrators to control which application components users can access and what operations they can perform.

User and Role Configuration

Four roles were defined in the Asset Lifecycle Management System:

1. Administrator: Responsible for overall application management, user administration, and access to all application records.
2. Object Manager: Responsible for registering objects, maintaining object information, and managing ownership transfers.

3. Inspector / Technician: Responsible for conducting inspections and handling maintenance-related records.

4. Approver: Responsible for reviewing ownership transfer requests and approving or rejecting them through the Blueprint process.

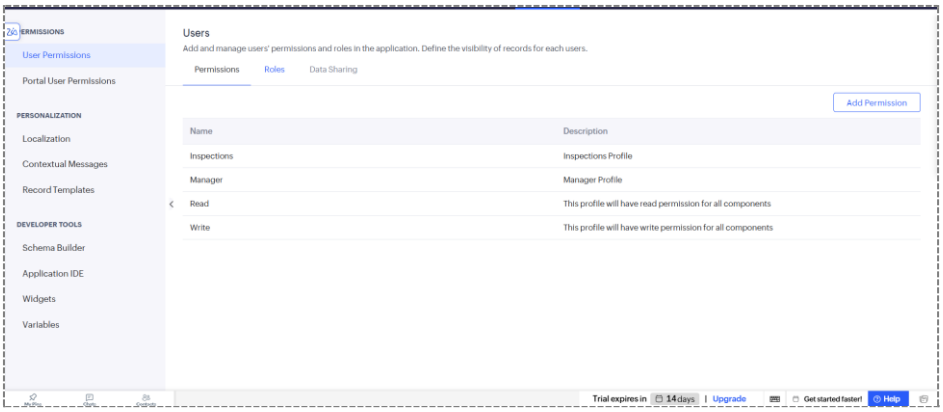


Figure 5.8: Roles and Permission Profiles Configuration

5.9 AI and Analytics Features

The Asset Lifecycle Management System incorporates Zoho Creator's built-in AI capabilities to simplify document processing and maintenance-related information analysis. Two AI features, Optical Character Recognition (OCR) and Keyword Extraction, were implemented to reduce manual effort and improve data handling.

Optical Character Recognition (OCR)

OCR is implemented in the Documents & Warranty form to extract text from uploaded documents such as warranty certificates and invoices.

- Users upload a document containing text.
- The OCR feature processes the uploaded document and recognises its text.
- The extracted text can be reviewed and used to retrieve relevant document information.
- This reduces the need for manually reading and entering information from documents.

Keyword Extraction

Keyword Extraction is implemented in the Maintenance Requests form to identify important terms from the issue description.

- Users enter a description of the maintenance problem.

- The AI feature analyses the description and extracts relevant keywords.
- For example, the description "Motor overheating with unusual vibration" may produce keywords such as motor, overheating, and vibration.
- These keywords help identify the main issue and make maintenance information easier to organise and search.

Analytics and Reporting

The application uses Zoho Creator's built-in reports and custom pages to organise and present asset-related information.

The nine reports cover objects, owners, documents and warranties, inspections, maintenance requests, ownership transfers, lifecycle history, service providers, and service records.

The application also includes four custom pages: Home Page, Admin Dashboard, Asset Dashboard, and Maintenance Centre. These provide access to the application's major functions and records.

AI Generation and Verification

The OCR and Keyword Extraction features use Zoho Creator's built-in AI capabilities. Their outputs were checked using sample warranty documents and maintenance descriptions to verify that relevant text and keywords were extracted.

The extracted results were reviewed for correctness before being used in the application. AI outputs were treated as assistance and not assumed to be error-free.

5.10 Pro-Code Extension

Pro-code development involves extending low-code applications using conventional programming languages, custom APIs, and external development platforms. It provides greater flexibility and control over application functionality.

The Asset Lifecycle Management System was developed primarily using Zoho Creator's low-code environment and Deluge scripting. Advanced pro-code extensions such as Zoho Catalyst, JavaScript widgets, and Creator REST API integrations were not implemented because the existing platform capabilities were sufficient to meet the project's requirements.

Potential Pro-Code Extensions

The application could be extended using the following technologies:

1. Zoho Catalyst: Could be used to develop independent backend services for advanced asset processing, scheduled operations, and scalable business logic.

2. JavaScript Widgets: Could provide customised user interfaces, interactive asset visualisations, and advanced dashboard components beyond the standard Page Builder capabilities.

3. Creator REST APIs: Could enable external applications to retrieve, create, update, and manage asset records securely through API requests.

4. Custom Functions: Additional reusable functions could be developed to handle complex calculations, validation, and business operations.

Justification

The project focused on implementing its core requirements using Zoho Creator's built-in features, workflows, Blueprint, and Deluge scripting. Introducing additional pro-code technologies would increase development complexity without being necessary for the current scope.

5.11 Mobile and Portal Access

The Asset Lifecycle Management System is developed using Zoho Creator, which provides access to applications through its mobile application for Android and iOS devices.

The application can be accessed through the Zoho Creator mobile app, allowing authorised users to work with asset-related information without depending exclusively on a desktop browser.

Mobile Accessibility

The following application features can be accessed through the mobile interface:

- Object Management: View and manage registered object information.
- Documents & Warranty: Access uploaded documents and warranty details.
- Maintenance Management: View maintenance requests and service records.
- Ownership Transfers: Access ownership transfer information and approval processes, subject to assigned permissions.
- Reports: View asset information and related reports.

Mobile-Specific Configuration

The application uses Zoho Creator's supported mobile interface. No separately developed native mobile application or custom mobile-specific layout was implemented. Offline access settings were also not specifically configured or tested.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing and debugging were performed to verify the functionality, reliability, and correctness of the Asset Lifecycle Management System. Unit testing was carried out on individual forms and Deluge scripts to check record creation, mandatory fields, lookup relationships, object age calculation, QR code generation, and automatic lifecycle history creation. Integration testing was performed to verify the interaction between object registration, ownership transfer, Blueprint approval stages, maintenance completion, and warranty reminder workflows. The four defined roles—Administrator, Object Manager, Inspector / Technician, and Approver—were considered during role-based access testing to ensure that access permissions matched their responsibilities. User Acceptance Testing (UAT) involved executing major application scenarios, such as registering objects, completing ownership transfers, recording maintenance activities, and accessing reports. Debugging was performed by identifying and correcting errors in scripts, field references, and workflow conditions. Sample records were used to verify the expected outcomes of the implemented functionalities

6.2 Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Register an object with valid details	Enter valid object name, category, purchase date, and owner. Submit the form.	Object record is created successfully.	Object record created successfully.	Pass
TC-02	FR-01	Register an object with missing mandatory information	Leave a mandatory field empty and submit the form.	Validation error is displayed and record is not created.	Validation error displayed; record creation prevented.	Pass
TC-03	FR-01	Verify automatic Object ID generation	Register a new object and inspect its Object ID.	A unique Object ID is generated automatically.	Unique Object ID generated automatically.	Pass

TC-04	FR-02	Verify object age calculation	Enter a valid purchase date and save the object.	Object Age is calculated automatically.	Object Age calculated automatically from Purchase Date.	Pass
TC-05	FR-02	Verify object age after changing purchase date	Update the purchase date and inspect Object Age.	Object Age reflects the updated purchase date.	Object Age updated according to the changed Purchase Date.	Pass
TC-06	FR-03	Verify registration history creation	Register a new object and open its Lifecycle History.	A registration event is automatically recorded.	Registration event automatically recorded in Lifecycle History.	Pass
TC-07	FR-03	Verify ownership transfer history	Complete an ownership transfer and inspect Lifecycle History.	An ownership transfer event is automatically recorded.	Ownership transfer event recorded in Lifecycle History.	Pass
TC-08	FR-04	Submit an ownership transfer request	Create a transfer request and submit it for review.	Request moves to the Under Review stage.	Transfer request successfully moved to Under Review.	Pass
TC-09	FR-04	Approve an ownership transfer	Select an Under Review request and approve it.	Request moves to the Approved stage.	Transfer request successfully moved to Approved.	Pass
TC-10	FR-04	Reject an ownership transfer	Select an Under Review request and reject it.	Request moves to the Rejected stage.	Transfer request successfully moved to Rejected.	Pass
TC-11	FR-04	Complete an approved transfer	Complete a request that has been approved.	Transfer status becomes Completed and the object's owner is updated.	Transfer status changed to Completed and object's owner updated.	Pass
TC-12	FR-04	Attempt an invalid transfer transition	Attempt to complete a transfer without completing the approval stage.	Blueprint prevents the invalid transition.	Invalid Blueprint transition prevented.	Pass

TC-13	FR-05	Verify Objects Report	Open the Objects Report after registering sample objects.	Registered object details are displayed correctly.	Objects Report displayed registered object details correctly.	Pass
TC-14	FR-05	Verify Ownership Transfers Report	Open the Ownership Transfers Report.	Ownership transfer records and their statuses are displayed.	Ownership Transfers Report displayed transfer records and statuses.	Pass
TC-15	FR-05	Verify Lifecycle History Report	Open the Lifecycle History Report.	Registration and ownership transfer events are displayed.	Lifecycle History Report displayed recorded events.	Pass
TC-16	FR-06	Verify warranty expiry reminder	Create a warranty record matching the configured expiry criteria and execute the scheduled workflow.	Eligible warranty records are identified and the configured reminder action is triggered.	Eligible warranty record identified and reminder action triggered.	Pass
TC-17	FR-06	Verify non-eligible warranty record	Create a warranty record that does not meet the expiry criteria.	No expiry reminder is triggered for that record.	Non-eligible warranty record did not trigger a reminder.	Pass
TC-18	FR-07	Verify service completion update	Change a service record's status to Completed.	The configured workflow updates the related object and lifecycle information.	Service completion workflow updated the related records.	Pass
TC-19	FR-07	Verify incomplete service record	Keep the service status as Pending and update the record.	Completion-related actions are not triggered.	Completion-related actions were not triggered for the pending service record.	Pass
TC-20	NFR-02	Test unauthorised access	Log in using a user account without permission to access a restricted component.	Access is denied according to the configured role permissions.	Unauthorised access to restricted components was prevented.	Pass

6.3 Debugging Deluge Scripts

Debugging was performed throughout the development of the Asset Lifecycle Management System to identify and resolve errors in Deluge scripts, form configurations, and automated workflows. Different debugging techniques were used to ensure that the application behaved as expected and that data was updated correctly.

The `info` statement in Deluge was used to display variable values and intermediate results during script execution. This helped identify incorrect values, verify record retrieval, and understand the execution flow. The Deluge editor's execution and debugging options were also used to check scripts and identify syntax or logical errors before using them in the application.

Workflow execution logs were examined to determine whether workflows were triggered correctly and whether their configured actions were executed. Error messages displayed during form submission were used to identify problems such as missing mandatory fields, incorrect field references, and invalid input values. These messages helped narrow down the source of errors and simplify troubleshooting.

One issue encountered during development involved the ownership transfer workflow, where the current owner of an object was not updated correctly after completing a transfer. The issue was traced to the lookup field reference used in the Deluge script. The field reference was corrected, and the workflow was tested again. Following the correction, the object's owner was updated successfully, and the corresponding ownership transfer event was recorded in Lifecycle History.

Repeated testing and debugging helped improve the reliability of the application. These techniques also ensured that important operations, such as object registration, ownership transfers, and maintenance updates, produced the expected results.

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Object's current owner was not updated after completing an ownership transfer.	Incorrect lookup field reference in the Deluge script.	Corrected the lookup field reference and retested the ownership transfer workflow.	Closed
BUG-02	Lifecycle history entry was not created as expected during initial script testing.	Incorrect field mapping in the history record insertion logic.	Corrected the field mappings and verified automatic history creation.	Closed

BUG-03	Object Age calculation produced an unexpected result during testing.	Incorrect handling of the date difference calculation.	Reviewed and corrected the formula used to calculate object age.	Closed

6.5 User Acceptance Testing

User	Role Tested	Feedback	Action Taken
User 1	Administrator	Application provides convenient access to object records and administrative functions.	Verified the administrative pages and reports.
User 2	Object Manager	Object registration and ownership management are easy to perform.	Verified object registration and ownership transfer workflows.
User 3	Inspector / Technician	Maintenance requests and service records are organised and accessible.	Verified maintenance forms and service completion workflow.
User 4	Approver	Ownership transfer stages make the approval process easier to manage.	Verified Blueprint transitions and ownership transfer completion.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The Asset Lifecycle Management System was developed using Zoho Creator. Forms, reports, pages, Deluge scripts, and workflows were configured and tested in the development environment. After verifying the major functionalities, the application was published and made accessible through its live application interface.

A separate Stage environment was not maintained. Testing was performed using sample records before making changes available to users. Version notes can be maintained to document updates and modifications to application components.

A rollback plan was considered to handle deployment issues. If an update causes unexpected errors, the previous stable version or configuration can be restored using available backup and version management options. This helps maintain application reliability and data integrity.

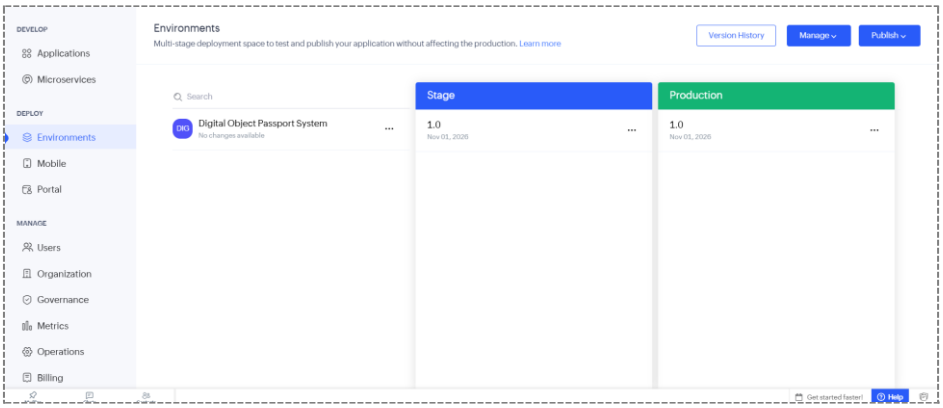


Figure 7.1: Environment

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	[ ]
2	Sample/test data removed or isolated	[ ]
3	Roles, permissions and users configured in Production	[ ]
4	Connections authorised in Production	[ ]

S.No.	Checklist Item	Done (✓)
5	Schedules and workflows enabled	[ ]
6	Application published and URL verified	[ ]
7	Version / release notes recorded	[ ]

7.2 Performance Optimisation

Performance optimisation was considered during the development of the Asset Lifecycle Management System to improve data retrieval and application responsiveness. Criteria-based record retrieval was used where applicable to access relevant records efficiently. Reports were organised to display relevant asset, ownership, maintenance, and lifecycle information without unnecessary data. Lookup relationships were used to connect related forms and maintain structured data. The application also avoided unnecessary duplication of records through automated lifecycle history updates. Since the project did not involve large-scale data processing, bulk operations and advanced query optimisation were not required. Performance was evaluated through normal application usage and report access; however, no formal response-time benchmarks or before-and-after measurements were conducted.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Criteria-based retrieval	Ownership Transfer	General record lookup	Specific record retrieval	Reduced unnecessary retrieval
Automated history creation	Object Registration	Manual entry	Automatic entry	Reduced manual effort
Filtered reports	Objects Report	All records displayed	Relevant records filtered	Easier data access

7.3 Security Features

Security was considered during the development of the Asset Lifecycle Management System to protect application data and control user access. Zoho Creator provides built-in authentication and permission management capabilities to support secure application access.

Four roles—Administrator, Object Manager, Inspector / Technician, and Approver—were created to separate responsibilities within the application. Role-based access helps ensure that users access application features according to their assigned responsibilities. The Ownership Transfer Blueprint provides an additional layer of control by requiring transfer requests to pass through defined approval stages before completion.

Mandatory field validation helps prevent incomplete or invalid data from being submitted. Zoho Creator's account authentication mechanism ensures that application access is restricted to authenticated users. These measures help protect asset information, maintain data consistency, and reduce the possibility of unauthorised ownership changes.

Record-level restrictions, encrypted fields, OAuth connections, IP restrictions, and portal security were not separately configured as part of this implementation. They can be considered for future improvements to strengthen application security further.

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role-based access control	Four application roles	Unauthorised access to application features
Blueprint approval	Ownership Transfers	Unauthorised ownership changes
Mandatory field validation	Application Forms	Invalid or incomplete data entry
Zoho account authentication	Application access	Unauthenticated access

7.4 Governance and Administration

Governance and administration practices were considered to maintain the Asset Lifecycle Management System and ensure its long-term reliability. Zoho Creator provides features for managing application changes, users, and stored data.

A consistent naming convention was followed for forms, fields, workflows, and reports to improve readability and simplify maintenance. Deluge scripts and workflows were reviewed and tested whenever modifications were made to avoid affecting existing functionality. Application records, including object registration and ownership transfer events, are maintained through the Lifecycle History form, providing a record of important asset activities.

Regular data backup and export can be performed using Zoho Creator's export facilities to support data recovery. User roles and access responsibilities are managed through the application's role configuration. The Administrator is responsible for managing users, reviewing application functionality, maintaining records, and making necessary updates. Periodic checks of workflows, reports, and application configurations help ensure that the system continues to operate correctly.

These practices support application maintainability, data consistency, accountability, and effective administration.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The Asset Lifecycle Management System was demonstrated using a realistic asset management scenario to verify the interaction between different application modules and user roles.

Scenario: A new piece of equipment is registered, inspected, maintained, and subsequently transferred to another owner. Its details and activities are recorded throughout its lifecycle.

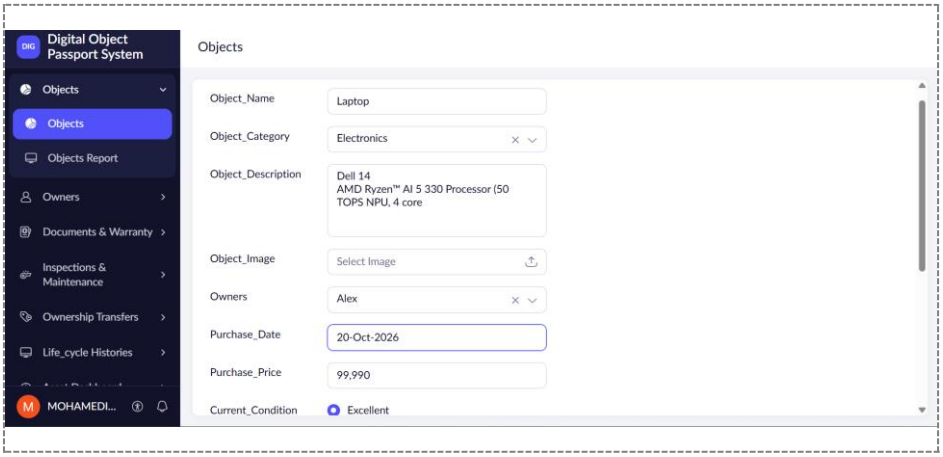


Figure 8.1: Scenario Step 1 – Object Registration

The Object Manager registers a new object by entering its name, category, purchase date, and owner details. The system generates an Object ID, calculates its age, and automatically creates a lifecycle history entry.

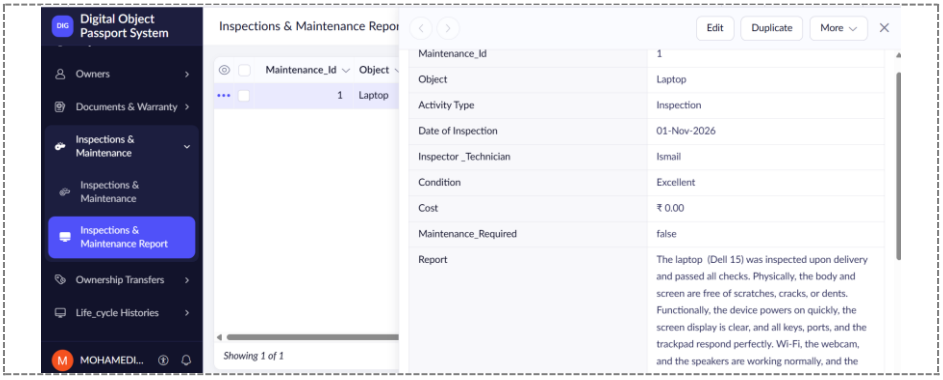


Figure 8.2: Scenario Step 2 – Inspection

The Inspector / Technician records the equipment's condition and inspection details. If maintenance is required, a maintenance request is created and the related service information is maintained.

8.2 Achievement of Objectives

Objective	Implemented Through	Outcome / Evidence
Develop a centralised asset management system.	Sections 5.2–5.4	Nine forms, nine reports, and four pages.
Automate asset management activities.	Sections 5.5–5.6	Lifecycle tracking, ownership updates, maintenance and warranty workflows.
Implement ownership approval.	Section 5.6	Ownership Transfer Blueprint with five stages.
Provide role-based access.	Section 5.8	Four application roles defined.
Integrate AI capabilities.	Section 5.9	OCR and Keyword Extraction implemented.
Test and deploy the application.	Sections 6.1–6.4	Application tested and published through Zoho Creator.

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Request Processing	Manual	Automated workflows	Reduced manual effort

Manual Steps	Multiple manual operations	Automated record updates	Fewer manual steps
Report Generation	Manual compilation	Built-in reports	Faster access to information
Data Accessibility	Scattered records	Centralised cloud application	Easier data access

8.4 Limitations of the Low-Code Approach

During the development of the Asset Lifecycle Management System, certain limitations of Zoho Creator were encountered. The Page Builder offered limited flexibility for creating highly customised user interfaces, so the application was designed using the available page components and built-in layouts. Deluge scripting also required adapting complex operations to the platform's supported syntax and execution environment. This was addressed by dividing application logic into smaller workflows and using lookup relationships to manage related records. Additionally, advanced integrations and external API capabilities were not explored due to the project's scope. Dependence on Zoho Creator also introduces vendor lock-in, as application logic and configurations rely on its platform-specific features. Despite these limitations, the built-in forms, reports, Blueprint, AI features, and Deluge scripting were sufficient to implement the required functionality without introducing unnecessary complexity.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The Asset Lifecycle Management System demonstrates how different levels of development can be combined using Zoho Creator. No-code configuration was used to create forms, reports, dashboards, lookup relationships, and user roles through visual tools. Low-code development was required for implementing Deluge scripts and workflows to automate object registration, lifecycle history creation, ownership transfers, maintenance updates, and warranty reminders. Built-in AI features such as OCR and Keyword Extraction were also configured through the platform.

Pro-code development was not required for the current project. However, technologies such as Zoho Catalyst, JavaScript widgets, and Creator REST APIs could be introduced

to develop advanced interfaces, independent backend services, and external integrations. Moving towards pro-code would become necessary if the application required complex business logic, extensive customisation, high-volume processing, or integration with multiple external systems. This project demonstrated that low-code development can efficiently address practical application requirements while retaining the flexibility to extend functionality through conventional programming when needed.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Asset Lifecycle Management System was successfully developed using Zoho Creator to provide a centralised platform for managing object information and tracking its lifecycle. The application includes nine forms, nine reports, and four custom pages to support object registration, ownership management, document and warranty tracking, inspections, and maintenance activities. Deluge scripting and workflows were implemented to automate lifecycle history creation, ownership updates, maintenance operations, and warranty monitoring. The Ownership Transfer Blueprint provides a structured approval process, while four user roles help organise application access and responsibilities.

Additionally, built-in AI features such as OCR and Keyword Extraction were integrated to simplify warranty document processing and maintenance request analysis. The application's major functionalities were tested using sample records and workflow scenarios before publishing it through Zoho Creator's live interface. Overall, the project demonstrates how low-code development, workflow automation, and AI capabilities can be combined to develop a practical and maintainable asset management solution.

9.2 Future Enhancements

The Asset Lifecycle Management System can be further enhanced with the following features to improve asset tracking, automation, and accessibility:

1. AI-Based Predictive Maintenance: Extend the existing maintenance module using Zia AI to analyse historical inspection and service records, identify patterns, and predict potential equipment failures.

2. Dynamic Digital Passport QR Code: Enhance the existing QR Code field to generate unique QR codes linked to individual digital passport pages. Scanning the code would allow authorised users to access object details, ownership history, warranty information, and maintenance records.

3. Advanced Warranty Document Processing: Extend the existing OCR functionality to automatically identify and extract important information such as warranty numbers, purchase dates, expiry dates, and manufacturer details from uploaded documents.

4. Maintenance Request Classification: Enhance Keyword Extraction to automatically categorise maintenance requests based on identified keywords and suggest their priority.

5. Interactive Asset Analytics: Improve the existing dashboards with visual charts showing asset distribution, condition status, maintenance frequency, warranty expiry trends, and ownership transfers.

6. Mobile-Based Inspection: Extend the application with mobile-friendly inspection features, allowing technicians to upload equipment photographs, update condition details, and record maintenance activities directly from the field.

7. External Integration: Integrate with external inventory or ERP systems through REST APIs to exchange asset information and reduce duplicate data entry.

9.3 Learning Outcomes and Career Relevance

Developing the Asset Lifecycle Management System provided practical experience in low-code application development using Zoho Creator. I gained knowledge of form creation, data modelling, reports, dashboards, Deluge scripting, workflow automation, Blueprint, role-based access control, and built-in AI features such as OCR and Keyword Extraction. I also gained experience in testing, debugging, application deployment, and understanding the limitations of low-code platforms.

These skills are relevant to several career pathways, including Low-Code Developer, Citizen Developer, Solutions Consultant, and Automation Engineer. The project helped me understand how business requirements can be converted into functional applications using visual development tools and scripting. It also provided a foundation for exploring pro-code technologies and integrating advanced capabilities into low-code applications.

REFERENCES

[1] A. Sahay, A. Indamutsa, D. Di Ruscio, and A. Pierantonio, "Supporting the understanding and comparison of low-code development platforms," in Proc. 46th Euromicro Conf. Software Engineering and Advanced Applications (SEAA), 2020, pp. 171–178, doi: 10.1109/SEAA51224.2020.00036.

[2] Zoho Corporation, "Zoho Creator Help Documentation." [Online]. Available: https://www.zoho.com/creator/help/. [Accessed: Nov. 1, 2026].

[3] Zoho Corporation, "Deluge Scripting Documentation." [Online]. Available: https://www.zoho.com/deluge/. [Accessed: Nov. 1, 2026].

[4] Zoho Corporation, "Creating Blueprints," Zoho Creator Help. [Online]. Available: https://help.zoho.com/portal/en/kb/creator/developer-guide/workflows/blueprints/articles/create-a-blueprint. [Accessed: Nov. 1, 2026].

[5] Zoho Corporation, "Understand Permissions," Zoho Creator Help. [Online]. Available: https://www.zoho.com/creator/newhelp/app-settings/understand-permissions.html. [Accessed: Nov. 1, 2026].

[6] Zoho Corporation, "AI Capabilities in Creator – Powered by Zia," Zoho Creator Help. [Online]. Available: https://help.zoho.com/portal/en/kb/creator/developer-guide/operations/zia/articles/ai-capabilities-in-creator-powered-by-zia. [Accessed: Nov. 1, 2026].

[7] Zoho Corporation, "Understand OCR Field," Zoho Creator Help. [Online]. Available: https://help.zoho.com/portal/en/kb/creator/developer-guide/forms/add-and-manage-fields/articles/fields-ocr-index. [Accessed: Nov. 1, 2026].

[8] Zoho Corporation, "Understand Keyword Extraction Field," Zoho Creator Help. [Online]. Available: https://www.zoho.com/creator/help/forms/aifields/keyword-extraction/. [Accessed: Nov. 1, 2026].

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Asset Lifecycle Management System	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Y	Nine application forms	5.2	
II	Lookup field (relationship between forms)	Y	Objects, Ownership Transfers, Lifecycle History	4.2, 5.2	
II	Subform (parent–child data)	Y	Inspections & Maintenance Form	4.2, 5.2	[ ]
II	Field validations / layout rules	Y	Application Forms	5.2	[ ]
II	List report with filter, grouping, custom action	Y	[ ]	5.3	[ ]
II	Kanban / Calendar / Timeline / Map report	Y	[ ]	5.3	[ ]
II	Pivot table / chart report	Y	[ ]	5.3	[ ]
II	Page / dashboard built with UI Builder	Y	Home Page, Admin Dashboard, Asset Dashboard, Maintenance Centre	5.4	[ ]
II	Deluge – form events (validate / success / field actions)	Y	Objects and Ownership Transfers workflow	5.5	[ ]
II	Deluge – functions, fetch, loops, collections	N	-	5.5	[ ]
II	Email / in-app notifications	Y	Maintenance Request Alerts	5.5, 5.6	[ ]

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Form workflow	Y	Object Registration History, Complete Ownership Transfer, Service Completion Update, Maintenance Request Alert	5.6	[ ]
II	Scheduled workflow	Y	Warranty Expiry Remainder	5.6	[ ]
II	Approval process	N	-	5.6	[ ]
II	Blueprint	Y	Ownership Transfer Approval	5.6	[ ]
II	Batch / report workflow	N	-	5.6	[ ]
II	Testing & debugging evidence	Y	Application forms, scripts and workflows	6	[ ]
III	Integration with a Zoho service	Y	Zoho Mail	5.7	[ ]
III	Connection / REST API call (invokeurl)	N	-	5.7	[ ]
III	Webhook or Creator REST API	N	-	5.7	[ ]
III	Users, roles and permission profiles	Y	Four roles and permission profiles created	5.8	[ ]
III	Record-level access criteria / customer portal	Y		5.8	[ ]
III	Zia / AI feature	Y	Documents & Warranty (OCR), Maintenance Requests (Keyword Extraction)	5.9	[ ]
III	Analytics (charts / Zoho Analytics)	Y	Reports and dashboards available	5.9	[ ]

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
IV	Performance optimisation applied & measured	Y	Basic criteria-based retrieval and filtered report access	7.2	[ ]
IV	Deployment across environments	Y	Deployment Environment	7.1	[ ]
IV	Security controls (encryption, OAuth, permissions)	N		7.3	[ ]
IV	Governance (audit trail, administration)	Y	Lifecycle History and Administrator role	7.4	[ ]
V	Pro-code extension (Catalyst / widgets / custom functions)	N	[ ]	5.10	[ ]
V	Low-code limitations & spectrum analysis	Y	Project limitations and low-code/pro-code reflection	8.4, 8.5	[ ]
—	Mobile app / portal access	Y	[ ]	5.11	[ ]

Total features implemented: 20 / 32

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	2.Medium	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	2.Medium	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	2.Medium	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	2.Medium	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	2.Medium	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 CompleteOwnership_Transfer

```
// Script ID : DS-01
// Location  : Ownership Transfers Form → Update of Transfer Status /
CompleteOwnership_Transfer]
// Purpose   : To complete the process of ownership transfer

if(input.Transfer_Status == "Completed")
{
if(input.Object != null && input.New__Owner != null)
{
objectRecord = Objects[ID == input.Object];
if(objectRecord.count() > 0)
{
objectRecord.Owners=input.New__Owner;
insert into Life_Cycle_History
[
Object=input.Object
Event_Type="Ownership Transfer"
Event_Date=zoho.currentdate
Event_Descr="Ownership transferred to a new owner."
Added_User=zoho.loginuser
]
}
}
}
```

C.2 Object Registration History

```
// Script ID : DS-02
// Location  : Objects Form -> On Successful Form Submission
// Purpose   : To Insert event into Lifecycle history

historyRecord = insert into Life_Cycle_History
[
Object=input.ID
Event_Type="Registered"
Event_Date=zoho.currentdate
Event_Descr="Digital passport created for " + input.Object_Name
Added_User=zoho.loginuser
];
```

C.3 Service Completion Update

```
// Script ID : DS-03
//Location : Service Records -> On Successful Form Submission
//Purpose: To Maintain records of the service
if(input.Service_Status == "Completed")
{
```

```
if(input.Objects != null)
{
objectRecord = Objects[ID == input.Objects];
if(objectRecord.count() > 0)
{
objectRecord.Current_Condition=input.Condition_After_Service;
}
insert into Life_Cycle_History
[
Object=input.Objects
Event_Type="Maintenance"
Event_Date=zoho.currentdate
Event_Descr="Maintenance completed. Work performed: " +
input.Work_Performed
Added_User=zoho.loginuser
]
}
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Web Access:

1. Open a web browser and navigate to the published Zoho Creator application link.
2. Log in using your registered Zoho account credentials.
3. Open the Asset Lifecycle Management System application.
4. Access the available forms, reports, and dashboards through the application menu.

Mobile Access:

1. Install and open the Zoho Creator mobile application.
2. Log in using your Zoho account.
3. Open the Asset Lifecycle Management System.
4. Navigate through the available application components.

D.2 Instructions for Administrator

- Log in and open the Admin Dashboard.
- Access the Objects form to view and manage registered objects.
- Manage owner, document, warranty, and service provider records.
- Monitor maintenance requests and ownership transfers.
- Open the Reports section to review all available reports.
- Monitor lifecycle history and scheduled workflow activities.

D.3 Instructions for Object Manager

- Log in and open the Asset Dashboard.
- Select the Objects form to register a new object.
- Enter object details, purchase date, category, and owner information.

- Open Documents & Warranty to upload relevant documents.
- Create ownership transfer requests when required.
- Use the Ownership Transfers Report to monitor transfer status.
- Access the Lifecycle History Report to review object activities.

D.4 Instructions for Inspector / Technician

- Log in and open the Maintenance Centre.
- Access the Inspections & Maintenance form.
- Select the relevant object and enter inspection details.
- Create or review maintenance requests.
- Open Service Records to record maintenance activities.
- Update the service status after completing maintenance.
- Access the Service Records and Maintenance Requests reports to review records.

D.5 Instructions for Approver

- Log in using the registered Zoho account.
- Open the Ownership Transfers form or its associated report.
- Select a transfer request that is in the Under Review stage.
- Review the object and new owner details.
- Select Approve Transfer or Reject Transfer.
- For approved requests, complete the transfer using the Blueprint transition.
- Verify the final status in the Ownership Transfers Report.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	Mohideen Abdul Kader ML	Register No.	240701329
Project Title	ASSET LIFECYCLE MANAGEMENT SYSTEM	Date	10-11-2026

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	Mohideen Abdul Kader ML	Register No.	240701329
Department / Section	CSE	Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER

Name: _____

EXTERNAL EXAMINER

Name: _____